Osmosis and Osmotic pressure of the solution

1. If 3 *gm* of glucose (*mol. wt.* 180) is dissolved in 60 *gm* of water at 15^oC. Then the osmotic pressure of this solution will be
(a) 0.34 atm (b) 0.65 atm
(c) 6.57 atm (d) 5.57 atm

2. The concentration in *gms* per *litre* of a solution of cane sugar $(M = 342)$ which is isotonic with a solution containing $6gms$ of urea $(M = 60)$ per *litre* is
(a) 3.42 (b) 34.2
(c) 5.7 (d) 19

3. Osmotic pressure is 0.0821 *atm* at temperature of $300K$. Find concentration in *mole/litre*
(a) 0.033 (b) 0.066
(c) 0.33×10^{-2} (d) 3

4. Osmotic pressure of a solution containing 0.1 mole of solute per litre at $273K$ is (in *atm*)
(a) $\frac{0.1}{1} \times 0.08205 \times 273$
(b) $0.1 \times 1 \times 0.08205 \times 273$
(c) $\frac{1}{0.1} \times 0.08205 \times 273$
(d) $\frac{0.1}{1} \times \frac{273}{0.08205}$

5. A solution contains non-volatile solute of molecular mass M_p. Which of the following can be used to calculate molecular mass of the solute in terms of osmotic pressure (*m* = Mass of solute, *V* = Volume of solution and π = Osmotic pressure)
(a) $Mp = \left(\frac{m}{\pi}\right)VRT$
(b) $Mp = \left(\frac{m}{V}\right)\frac{RT}{\pi}$
(c) $Mp = \left(\frac{m}{V}\right)\frac{\pi}{RT}$
(d) $Mp = \left(\frac{m}{V}\right)\pi RT$

6. The osmotic pressure of a 5% (*wt/vol*) solution of cane sugar at 150^oC is
(a) 2.45 *atm* (b) 5.078 *atm*
(c) 3.4 *atm* (d) 4 *atm*

7. The relationship between osmotic pressure at $273K$ when $10g$ glucose (P_1),$10g$ urea (P_2) and $10g$ sucrose (P_3) are dissolved in $250ml$ of water is
(a) $P_1 > P_2 > P_3$
(b) $P_3 > P_1 > P_2$
(c) $P_2 > P_1 > P_3$

(d) $P_2 > P_3 > P_1$

8. In osmosis
(a) Solvent molecules move from higher concentration to lower concentration
(b) Solvent molecules move from lower to higher concentration
(c) Solute molecules move from higher to lower concentration
(d) Solute molecules move from lower to higher concentration

9. Semipermeable membrane is that which permits the passage of
(a) Solute molecules only
(b) Solvent molecules only
(c) Solute and solvent molecules both
(d) Neither solute nor solvent molecules

10. Two solutions *A* and *B* are separated by semi- permeable membrane. If liquid flows form *A* to *B* then
(a) *A* is less concentrated than *B*
(b) *A* is more concentrated than *B*
(c) Both have same concentration
(d) None of these

11. A 5% solution of canesugar (mol. wt. =342) is isotonic with 1% solution of a substance X . The molecular weight of X is
(a) 34.2 (b) 171.2
(c) 68.4 (d) 136.8

12. Which of the following colligative properties can provide molar mass of proteins (or polymers or colloids) with greater precision
(a) Relative lowering of vapour pressure
(b) Elevation of boiling point
(c) Depression in freezing point
(d) Osmotic pressure

13. The average osmotic pressure of human blood is 7.8 bar at 37^oC. What is the concentration of an aqueous $NaCl$ solution that could be used in the blood stream
(a) $0.16 mol/L$ (b) $0.32 mol/L$
(c) $0.60 mol/L$ (d) $0.45 mol/L$

14. A solution of sucrose(molar mass = 342 *g/mol*) is prepared by dissolving 68.4 *g* of it per *litre* of the solution, what is its osmotic pressure (*R* = 0.082 *lit. atm.*$k^{-1}mol^{-1}$) at 273*k*
(a) 6.02 *atm* (b) 4.92 *atm*
(c) 4.04 *atm* (d) 5.32 *atm*

15. Blood has been found to be isotonic with
(a) Normal saline solution
(b) Saturated $NaCl$solution
(c) Saturated KCl solution
(d) Saturated solution of a 1 : 1 mixture of $NaCl$ and KCl

16. If 20 *g* of a solute was dissolved in 500 *ml* of water and osmotic pressure of the solution was found to be 600 *mm* of *Hg* at 15^oC, then molecular weight of the solute is
(a) 1000 (b) 1200
(c) 1400 (d) 1800

17. The osmotic pressure of 0.4% urea solution is 1.66 atm and. that of a solution of suger of 3.42 % is 2.46 atm. When both the solution are mixed then the osmotic pressure of the resultant solution will be
(a) 1.64 atm (b) 2.46 atm
(c) 2.06 atm (d) 0.82 atm

18. Blood is isotonic with
(a) 0.16 *M* $NaCl$
(b) Conc. $NaCl$
(c) 50 % $NaCl$
(d) 30 % $NaCl$

19. Which inorganic precipitate acts as semipermeable membrane **or** The chemical composition of semipermeable membrane is
(a) Calcium sulphate
(b) Barium oxalate
(c) Nickel phosphate
(d) Copper ferrocyanide

20. The osmotic pressure of $1m$ solution at 27^oC is
(a) 2.46 *atm* (b) 24.6 *atm*
(c) 1.21 *atm* (d) 12.1 *atm*